

0 Probability and Statistics

0.1 Probability Theory

In probability theory we imagine running an **experiment** which can have a number of **outcomes**. An **event** is a particular outcome, or set of outcomes. For example, in a dice roll, the outcomes are 1,2,3,4,5,6 and some events are getting a five {5} or rolling an even number {2, 4, 6}. A probability function is a way of associating a value with every possible event. A probability function must satisfy the following properties

1. $P(\Omega) = 1$
2. For any event E in Ω , $P(E) \geq 0$
3. If A and B are **disjoint** events then $P(A \cup B) = P(A) + P(B)$

where Ω is the set of all possible events. You can think about probability like the area in a Venn diagram

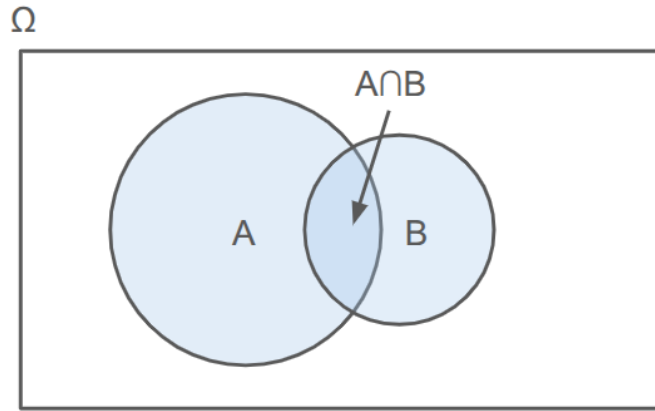


Figure 1: Venn diagram

The figure above shows that when events A and B , overlap the probability of A or B , $P(A \cup B)$, is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

we have to subtract off the intersection to avoid double counting.

The **conditional probability** is the probability that A happened given that B already happened. This is how we can update probabilities in light of new information. We have the formula

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

for $P(A|B)$, the probability of A *given* B . Again, referring to Figure 1 knowing that B happened restricts the sample space down from Ω to just B . Then the chance that A happened is just the fraction of B that overlaps with A .

If knowing B has no effect on the probability of A then A and B are said to be **independent**. We write this as

$$\begin{aligned} P(A) &= P(A|B) = \frac{P(A \cap B)}{P(B)} \\ \implies P(A \cap B) &= P(A)P(B) \end{aligned}$$

i.e. for independent events the probabilities multiply

A **random variable** (rv) is (despite the name) a *function* that associates a number with every event. Flipping a coin twice gives the following sample space

$$\{HH, TH, HT, TT\}$$

Some random variables are: the number of heads, the number of tails, the square root of the number of tails etc. This is an example of a **discrete random variable**, where only a finite set of outcomes are possible. A **continuous random variable** would be something like the height of a random person which takes a real number value. We will often approximate discrete rvs with continuous ones, since they are often easier to work with.

The **cumulative distribution function** (cdf) is

$$F(x) = P(X \leq x)$$

The **probability density function** (pdf) is

$$f(x) = P(X = x)$$

the probability of the rv X taking the value x .

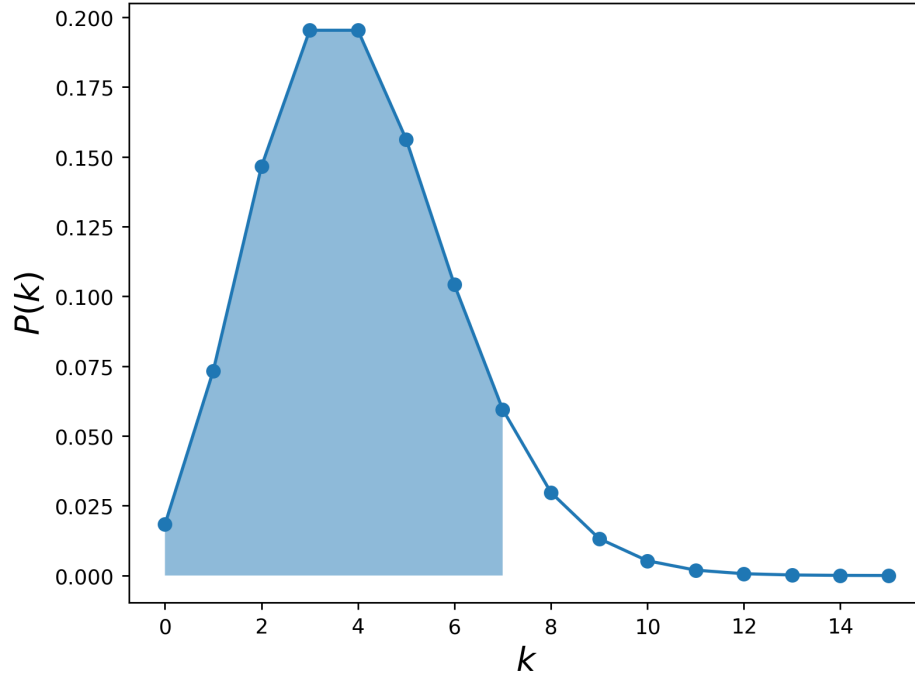


Figure 2: A discrete probability distribution

Figure 2 above shows a probability distribution. The value of the pdf at x gives the probability of observing the value x . The cdf at x is given by the area to the left of x , in the figure the area shaded in blue is $F(7)$.

Some useful formulas relating to pdfs and cdfs are

$$F(-\infty) = 0$$

$$F(\infty) = 1$$

We also have

$$P(X > x) = 1 - F(x)$$

which is the area under the pdf to the right of x .

$$P(a \leq X \leq b) = F(b) - F(a)$$

if the area between a and b . By taking the limit of this formula as the interval gets very small we get

$$f(x) = \frac{dF}{dx}$$

The simplest example of a probability distribution is a **Bernoulli distribution**. We imagine flipping a biased coin which lands heads with probability p and otherwise tails with probability $q = 1 - p$. The random variable X takes the value 1 (success) for heads and 0 (failure) for tails. So

$$f(0) = 1 - p \quad , \quad f(1) = p$$

$$F(0) = 1 - p \quad , \quad F(1) = 1$$

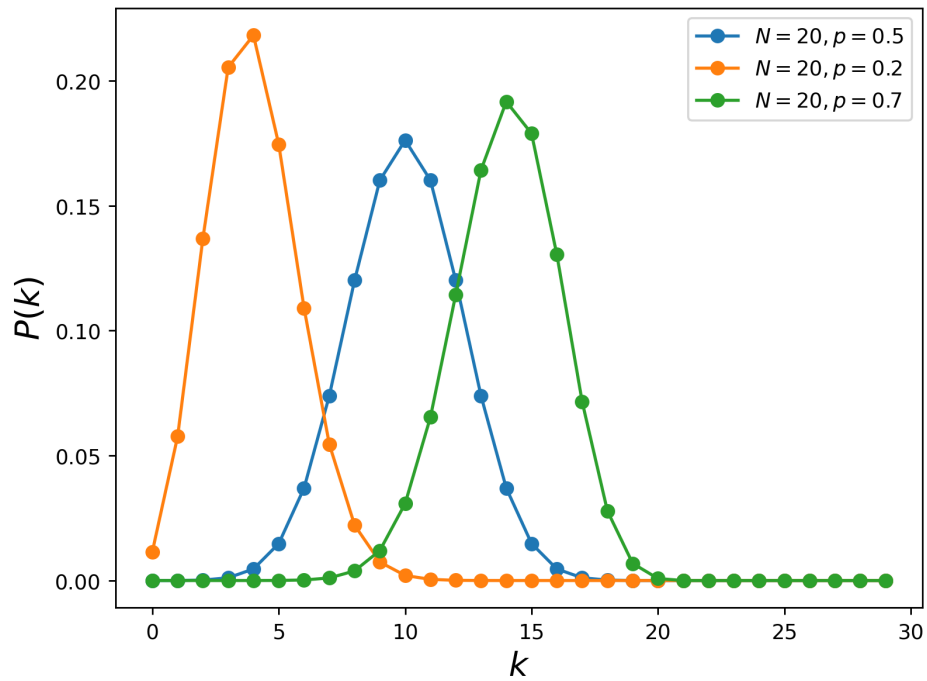


Figure 3: Some example binomial distributions

Now imagine doing multiple independent flips with the random variable being the number of heads. To get k heads in N flips we need to succeed k times, which happens with probability p^k , and fail $N - k$ times, which

happens with probability $(1 - p)^{N-k}$. These are independent so we can multiply the probabilities. We also have to account for the many different ways to get the same number. For example *HHTTH* and *HTHTH* both have three heads. There are

$$\binom{N}{k} = \frac{N!}{k!(N-k)!}$$

ways to to put k heads into N positions. So

$$f(k) = \binom{N}{k} p^k (1-p)^{N-k}$$

this is the **binomial distribution**

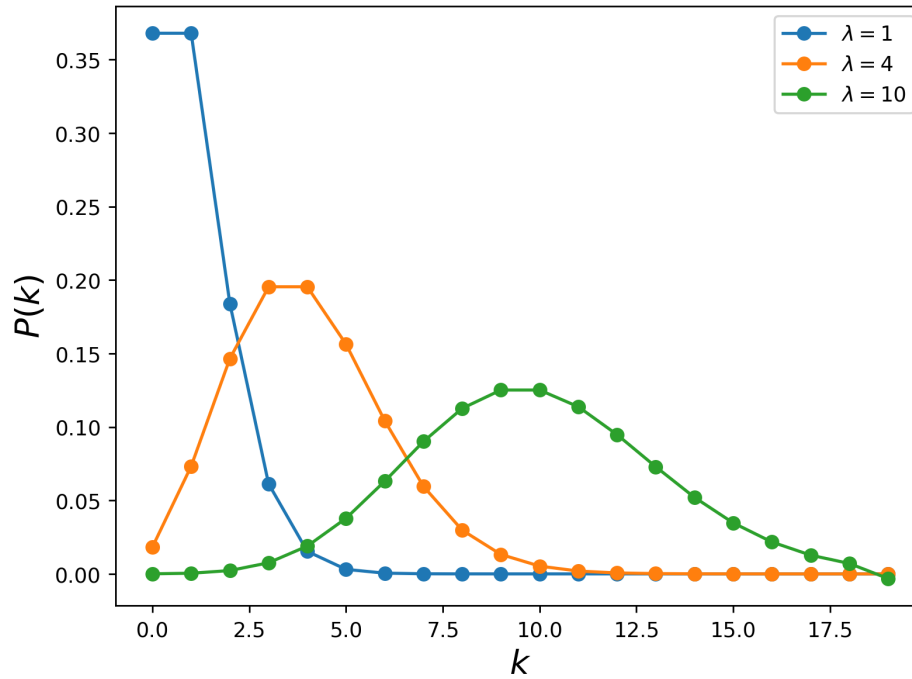


Figure 4: Some example Poisson distributions

A related distribution is the **Poisson distribution** which says the probability of observing the (integer) k is

$$f(k; \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$$

here λ is a **parameter** which controls the shape of the distribution. We can show that for certain large values of N this approximates the binomial distribution. But we can also simply observe that many real data sets (radioactive decay, buses arriving at a bus stop and so on) can be very accurately modelled by this kind of curve.

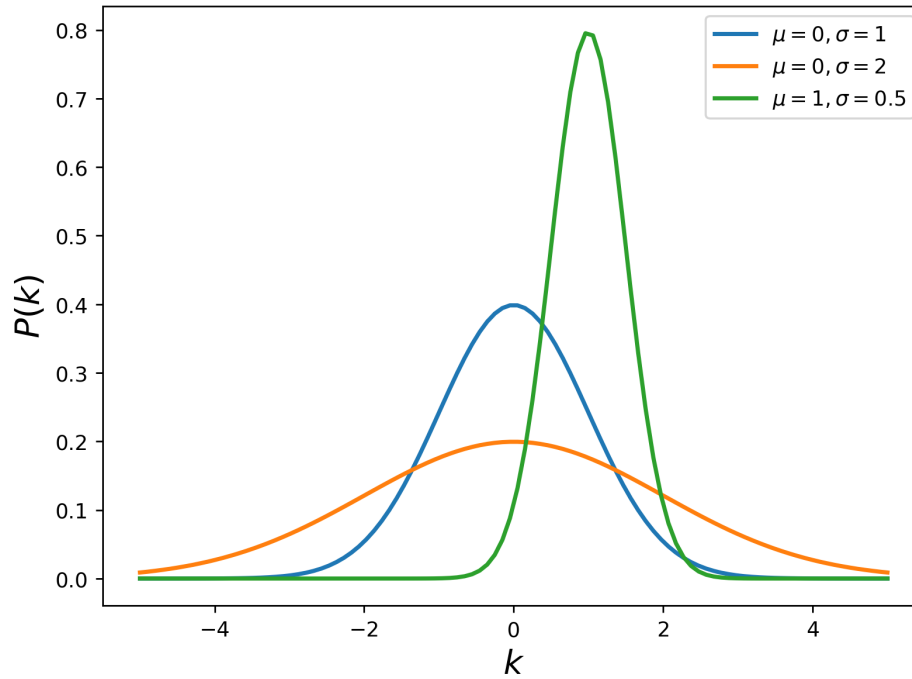


Figure 5: Some example normal distributions

The most common distribution is the **Gaussian distribution** also called the **normal distribution**.

$$f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

We will show later that this is the distribution governing sample averages, and many real data sets are normally distributed.

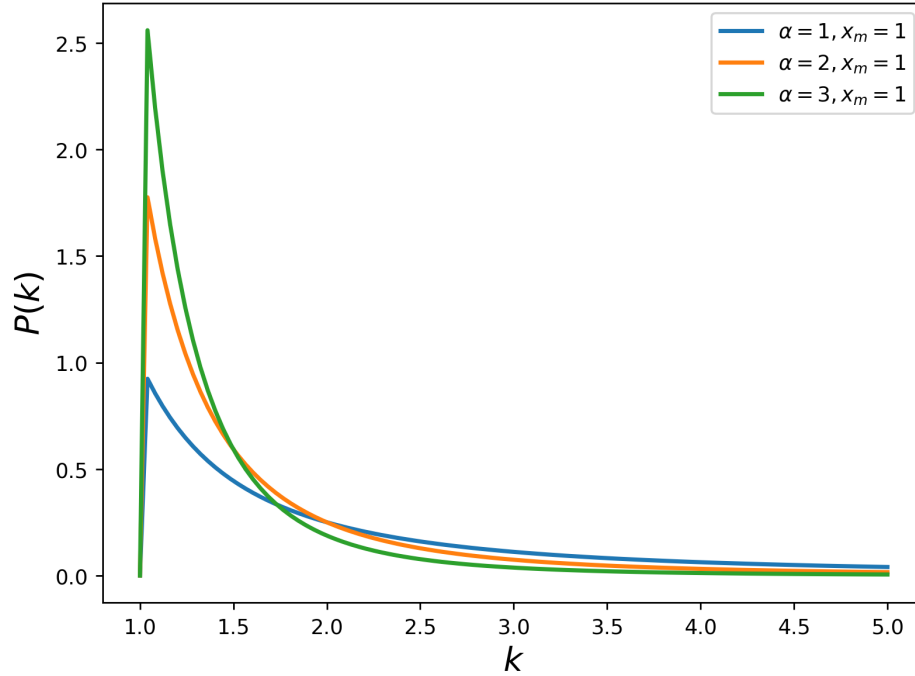


Figure 6: Some example Pareto distributions

In this course we frequently encounter **power law distributions** also called **Pareto distributions**

$$P(X > x) = 1 - F(x; \alpha, x_m) = \begin{cases} \left(\frac{x_m}{x}\right)^\alpha & x \geq x_m \\ 1 & x < x_m \end{cases}$$

where x_m is the minimum value that can be observed and α is some exponent. The pdf is

$$f(x; \alpha, x_m) = \begin{cases} \frac{\alpha x_m^\alpha}{x^{\alpha+1}} & x \geq x_m \\ 0 & x < x_m \end{cases}$$

The thing to note is that, although the probability decreases with x it does so *much* more slowly than exponentially, so the probability of large values is much higher than for the Poisson or Gaussian distributions.

0.2 Expectations

The **expected value** of a discrete random variable is

$$E[X] = \sum_i x_i P(x_i)$$

where the sum ranges over all possible events. For a continuous random variable we replace the sum with an integral

$$E[X] = \int_{-\infty}^{\infty} x f(x)$$

The expected value is also called the **mean** and gives a sense of the most likely value to be observed. However there are other ways to measure this, the two most common are the **median** m which is the value of x that ‘splits’ the probability in half, so we are just as likely to measure a value higher or lower than m

$$F(X = m) = 0.5$$

Another way to summarise a distribution is the **mode** ν which is the value of x at which the probability is maximised

$$\nu = \operatorname{argmax}(f(X = x))$$

In general

$$\mu \neq m \neq \nu$$

The mean μ is easiest to work with mathematically, while the median is quite robust and useful for real data sets. The mode is rarely useful.

A measure of the ‘spread’ of a probability distribution is its **variance**

$$\sigma^2 = V[X] = E[(X - \mu)^2]$$

Using the formula for expectation this can be rewritten as

$$V[X] = E[X^2] - (E[X])^2$$

The square root of the variance σ is called the **standard deviation**. The table shows the means and variances for the distributions we have introduced

Distribution	Mean	Variance
Bernoulli	p	$p(1 - p)$
Binomial	np	$np(1 - p)$
Poisson	λ	λ
Gaussian	μ	σ^2
Pareto	$\begin{cases} \infty & \alpha \leq 1 \\ \frac{\alpha x_m}{\alpha - 1} & \alpha > 1 \end{cases}$	$\begin{cases} \infty & \alpha \leq 2 \\ \frac{\alpha x_m^2}{(\alpha - 1)^2 (\alpha - 2)} & \alpha > 2 \end{cases}$

A very important result is the **Law of Large Numbers** which states: For a sequence of independent, identically distributed (iid) random variables $X_1, X_2, \dots$, the **sample mean**

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

gets closer and closer to the true mean μ . Consider a coin tossing experiment. The mean of the Bernoulli distribution is p . The law of large numbers says that as we do many coin tosses the sample mean, which is just the sum of the outcomes divided by the number of trials, approaches the theoretical mean, in this case p . This means that measuring a lot of data can give us information about the underlying probability distribution, in particular, its mean.

The **central limit theorem** goes further and tells us for some fixed number of trials n how far away are we from the theoretical expectation. It says for a sequence of iid random variables the sample mean $\bar{X}_n$ follows a normal distribution. More precisely it says that

$$(\bar{X}_n - \mu)\sqrt{n} \sim \mathcal{N}(0, \sigma^2)$$

where μ and σ^2 are the mean and variance of the Gaussian and the notation $\sim \mathcal{N}(0, \sigma^2)$ means ‘follows a normal distribution with mean 0 and variance σ^2 ’.

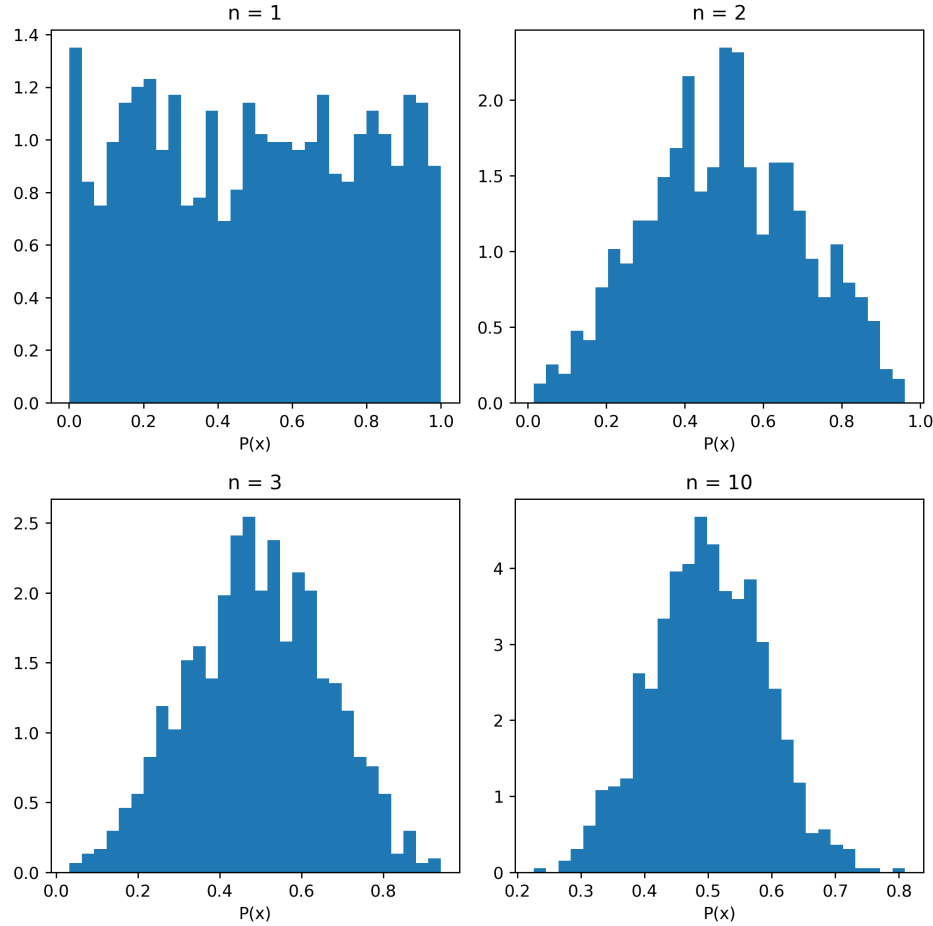


Figure 7: The plots show the histogram we get by calculating many sample averages using n data points sampled from a **uniform distribution** (constant probability of any number between 0 and 1). The distributions get more and more normal as n increases.

0.3 Hypothesis Testing

A common way to test scientific hypotheses like ‘does drug X cure cancer’ is statistical hypothesis testing. First we state a **null hypothesis**, H_0 , which gives the default position e.g. ‘the drug has no effect’. The **alternative hypothesis** H_a is the logical opposite of the null $H_a = !H_0$ i.e. ‘the drug has an effect’. The goal of hypothesis testing is to determine how plausible

the null hypothesis is. So the outcome is something either reject H_0 : ‘it is implausible that the drug has no effect’ or failing to reject H_0 : ‘it is *not* implausible that the drug has no effect’. To carry this out in practice we need to understand the probability distribution specified by the null and then quantify what we mean by ‘implausible’.

We first construct a **test statistic**, which is some number associated with the outcome of the experiment. When we have a test statistic we compute the **p-value**, which is the probability that a test statistic as extreme as the one that was observed would occur if the null hypothesis were true. Usually we are hoping for *small* p-values, they indicate that the null hypothesis is likely false, and that something interesting has happened!

The p-value is the probability of seeing a deviation as large or larger than the one observed. For test statistic T and observed value t the right tail p-value is

$$p_r = P(T \geq t | H_0)$$

The left tail is

$$p_l = P(T \leq t | H_0)$$

and the two-sided p-value is

$$p = 2 \min(p_l, p_r)$$

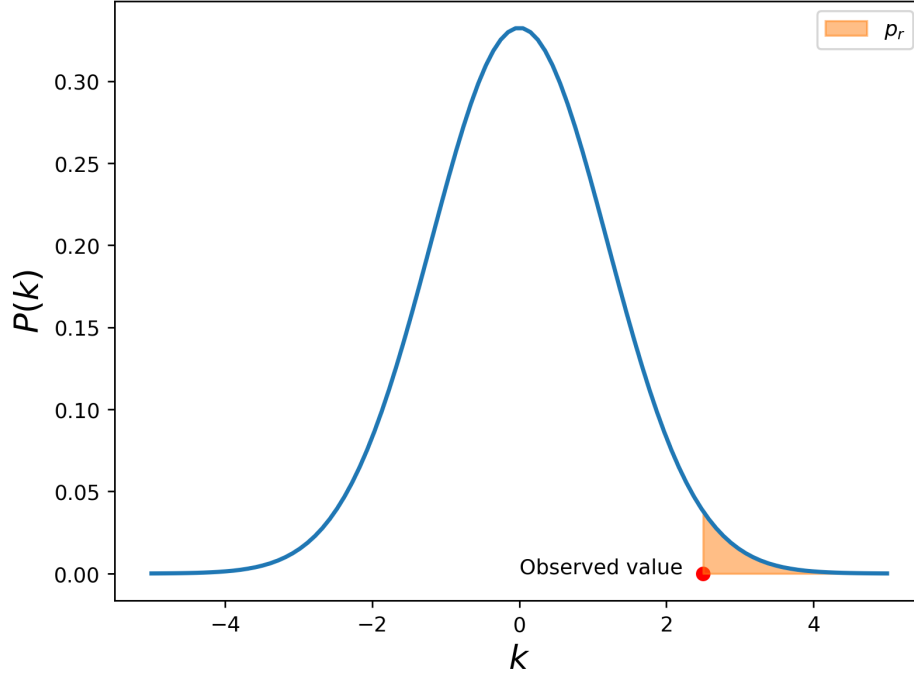


Figure 8: The observation and the corresponding right tailed p -value.

Depending on our experiment we might use any of these. The **significance level** α is some number specified ahead of time which defines what we mean by ‘implausible’. For example if we choose $\alpha = 0.01$ we reject the null if the p -value is less than this. This amounts to saying that if our observed test statistic only occurs by chance once in one hundred experiments, we are happy to rule out the null and declare an effect.

Here is a simple example: we flip a coin a number of times. The null hypothesis is that the coin is fair (lands heads or tails with equal probability). The test statistic, X , is the number of heads. Let’s suppose we do 5 flips and get five heads, $X = 5$. The probability of this under the null hypothesis is $(1/2)^5 \sim 0.03$. In this case the (right tailed) p -value is 0.03. If we do 10 flips and get all heads the p -value is $(1/2)^{10} \sim 0.001$. The **power** of an experiment is the probability of detecting an effect in the test. So the experiment with 10 flips has more power than the 5 flip case, because we can determine with more confidence if the coin is biased or not.

There are two main ways to be wrong with significance testing.

1. **Type 1 error** also called a **false positive** where we reject a true null

hypothesis.

2. **Type 2 error** also called a **false negative** where we fail to reject a false null hypothesis.

To help remember think of a medical test, a false positive means the test says you have the disease but you don't, while a false negative is when the test says you are healthy but you are not!

0.4 Likelihood

As we have seen, probability distributions are often described by parameters. If we believe our observed data is sampled from a particular probability distribution then we would like a way to estimate those parameters. We have seen how the mean can be estimated by the sample average, but things aren't always so simple, for example, how can we best estimate the exponent and minimum of the power-law distribution?

One way to approach this is **the method of maximum likelihood**. The **likelihood** is just the pdf, but instead of thinking of it as a function of x , the values of the random variable, we think of it as a function of parameters

$$\mathcal{L}(\theta; \vec{y}) = f(\vec{y}|\theta)$$

with $\vec{y}$ is some fixed vector of observations. The **maximum likelihood estimator** of some parameter θ is the solution to

$$\hat{\theta} = \operatorname{argmax}_{\theta} \mathcal{L}(\theta; \vec{y})$$

For this distribution, $\theta = \hat{\theta}$ this makes the probability of generating the observed data as high as possible.

Because many probability distributions are written in terms of exponentials and powers, it is often simpler to deal with the **log likelihood**

$$\log \mathcal{L}$$

maximising this gives the same result as maximising the likelihood, but is often simpler. Likelihood values are often very small, so working with the logarithm also helps avoid numerical issues.

If the data are iid and each is sampled independently from the same probability distribution f then we can multiply the probabilities

$$\mathcal{L}(\theta; \vec{y}) = \prod_i f(\vec{y}_i|\theta)$$

taking the log

$$\log \mathcal{L}(\theta; \vec{y}) = \sum_i \log f(\vec{y}_i | \theta)$$

We can use this to get an estimate of the exponent of a power law. An alternative way to write the power law distribution is (send $\alpha \rightarrow \alpha + 1$)

$$f(x; \alpha, x_m) = \frac{\alpha - 1}{x_m} \left(\frac{x}{x_m} \right)^{-\alpha}$$

If all the data are independent then the log likelihood is

$$\begin{aligned} \log \mathcal{L} &= \sum_i \log f(x_i; \alpha, x_m) \\ &= \sum_i \log \left(\frac{\alpha - 1}{x_m} \left(\frac{x_i}{x_m} \right)^{-\alpha} \right) \\ &= \sum_i \log(\alpha - 1) - \log x_m - \alpha \log \frac{x_i}{x_m} \\ &= n \log(\alpha - 1) - n \log x_m - \alpha \sum_i \log \frac{x_i}{x_m} \end{aligned}$$

We maximise this as usual by differentiating it and setting the derivative to zero

$$\begin{aligned} \frac{d \log \mathcal{L}}{d\alpha} &= 0 \\ \frac{n}{\hat{\alpha} - 1} - 0 - \sum_i \log \frac{x_i}{x_m} &= 0 \end{aligned}$$

Solving gives

$$\hat{\alpha} = 1 + n \left(\sum_i \log \frac{x_i}{x_m} \right)^{-1}$$

This is a robust way to estimate the exponent of a power law distribution, compared to say, fitting a straight line to the log-log plot.

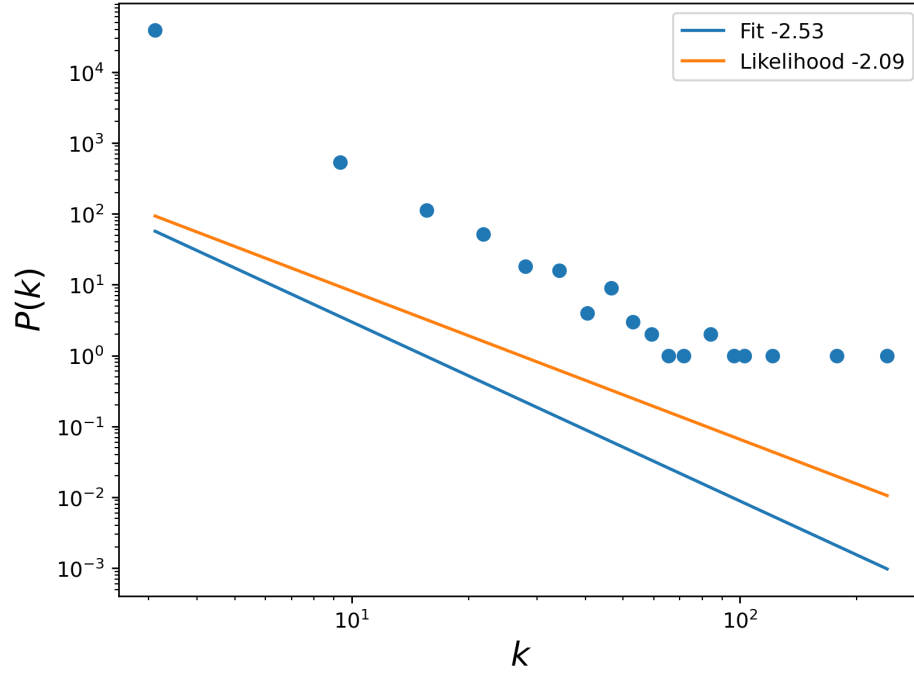


Figure 9: Data sampled from a power law. The slope is obtained in two ways, fitting the scatter and using the likelihood formula. The true value is -2 , showing the likelihood method is much closer.